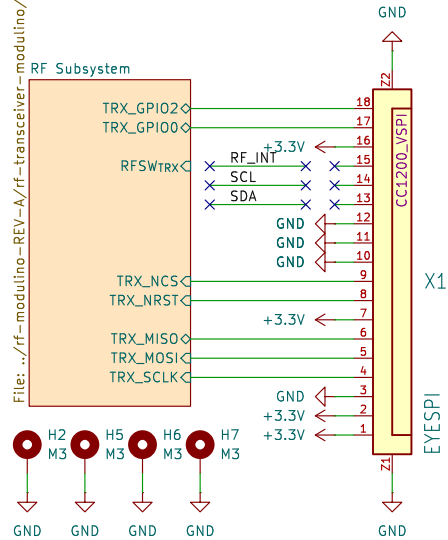
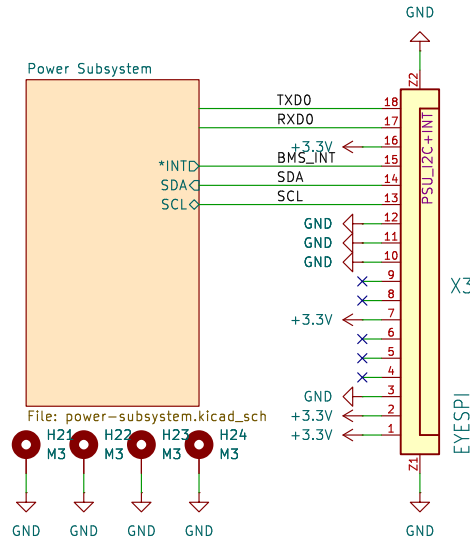


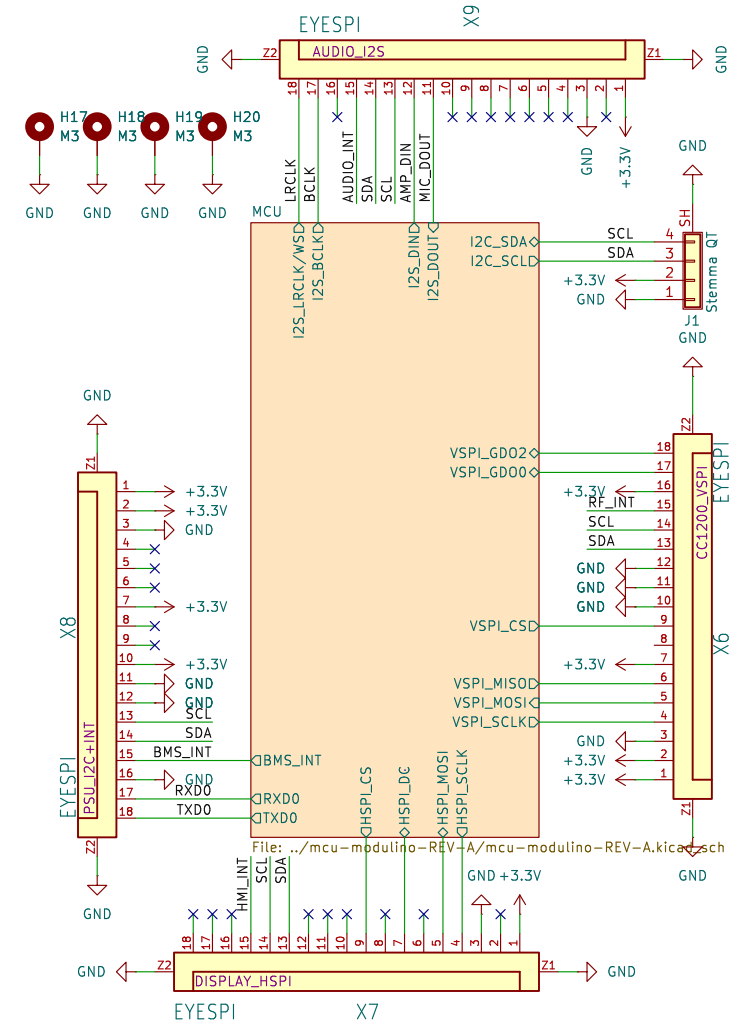
RF Modulino
 CC1200 on VSP1
 Unused I2C0+INT
 RX switch on ???
 4x +3.3V return paths



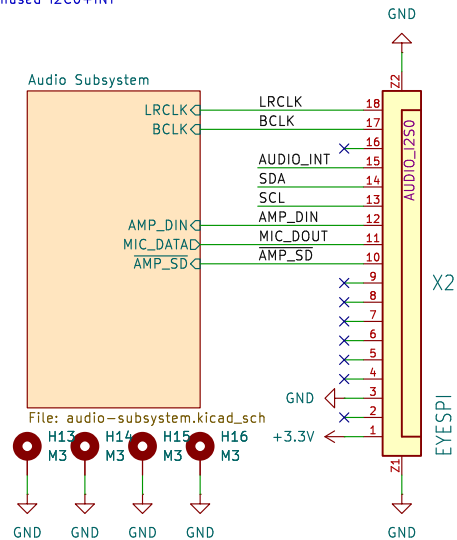
Battery Modulino
 BMS on I2C0 @ 0x6B



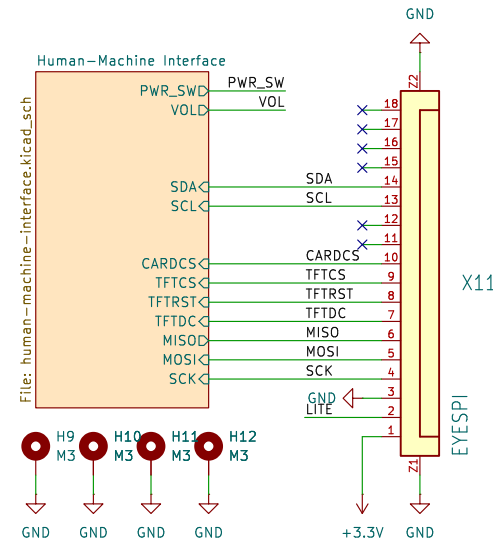
ESP32 Modulino
 X? EYESPI to Modulino Nodes



Audio Modulino
 Mic+AMP on I2S0:
 MAX98357A I036 INPUT
 ICS-43434 I033 OUTPUT
 Unused I2C0+INT



Human-Machine Interface Modulino
 Display on HSPI:
 SD Card Unused
 Port Expander on I2C0+INT @ ???



Sheet: /
 File: system-testbed.kicad_sch

Title:

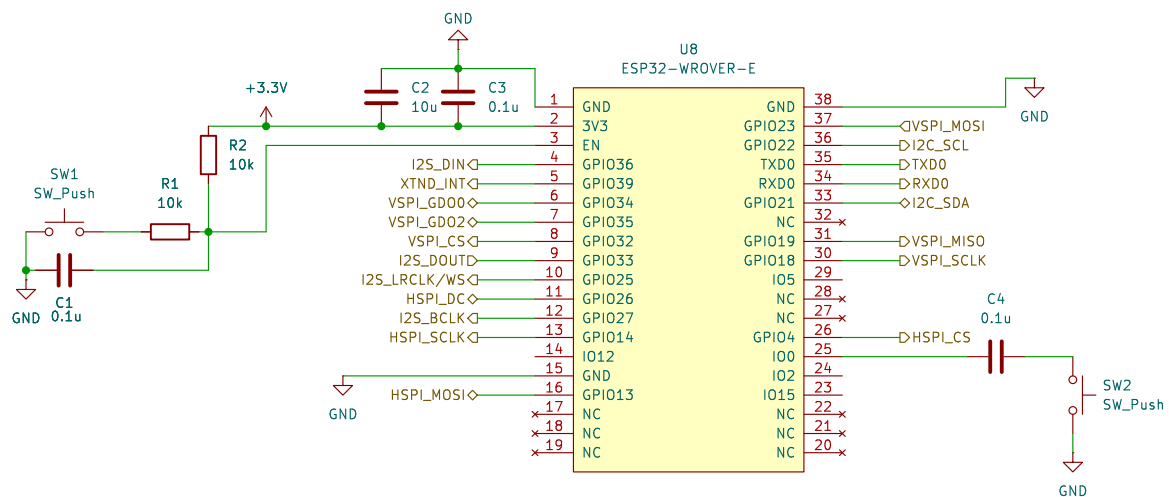
Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 1/12



Sheet: /MCU/
File: mcu-modulino-REV-A.kicad_sch

Title:

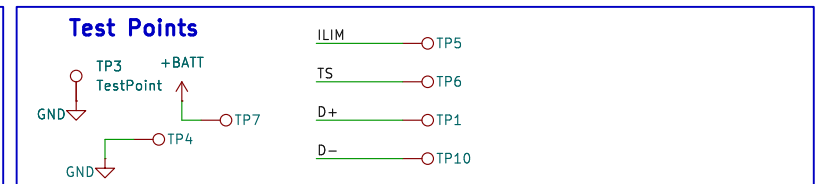
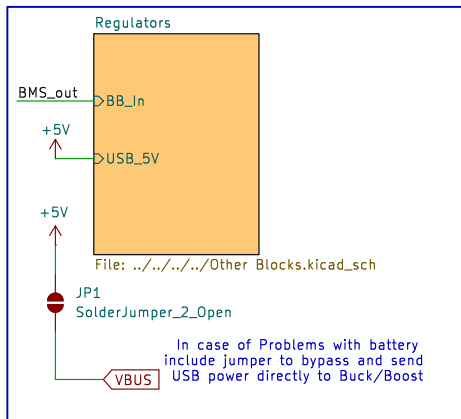
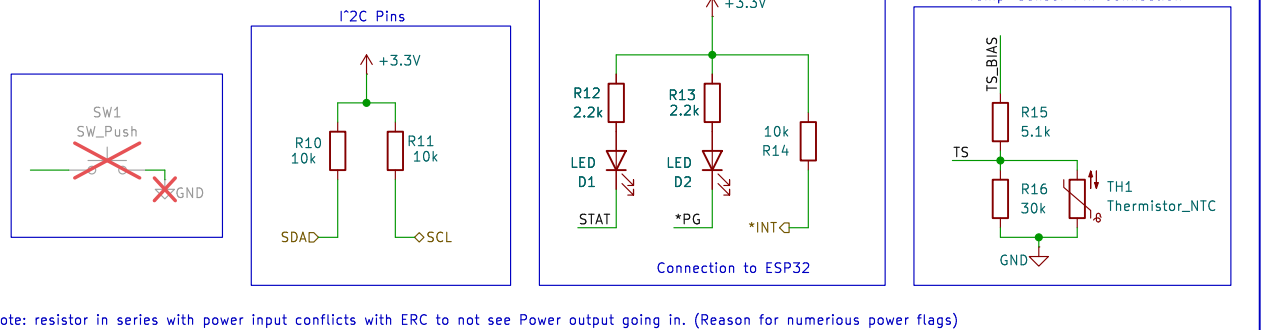
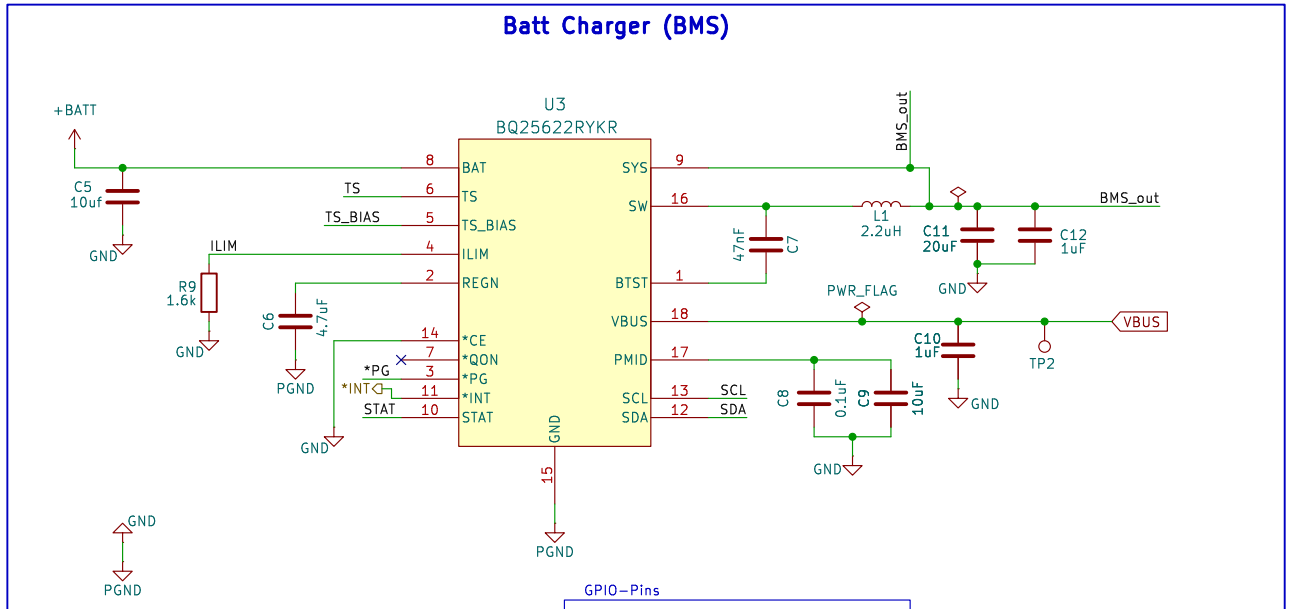
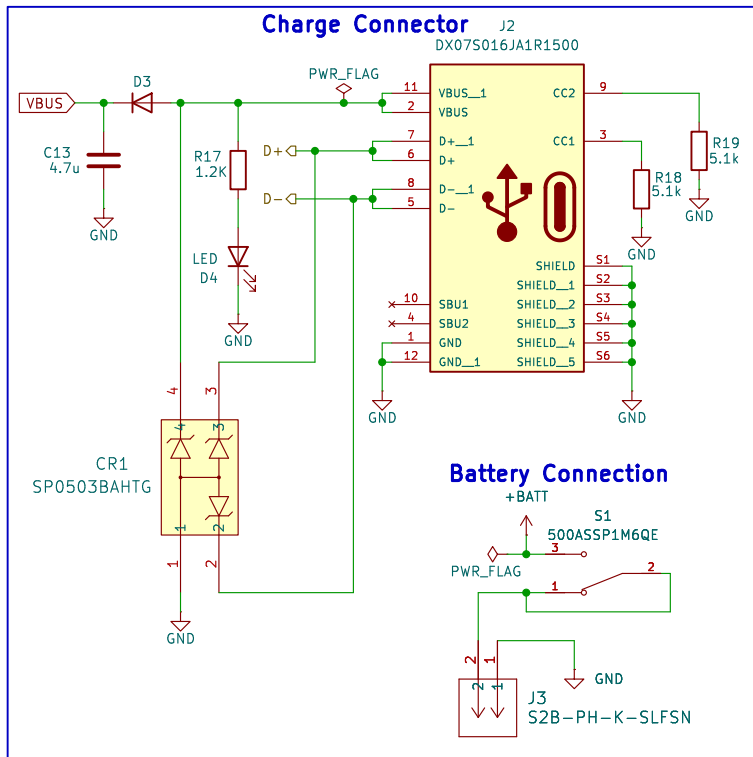
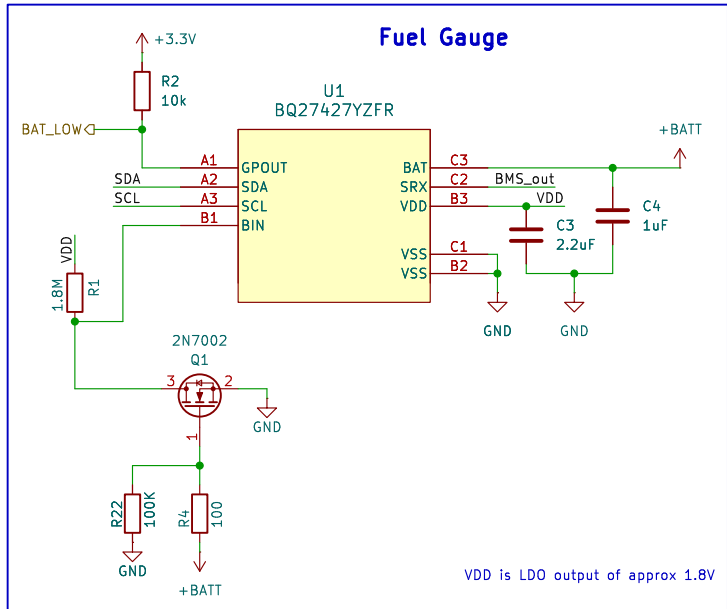
Size: A4

Date:

KiCad E.D.A. 9.0.6

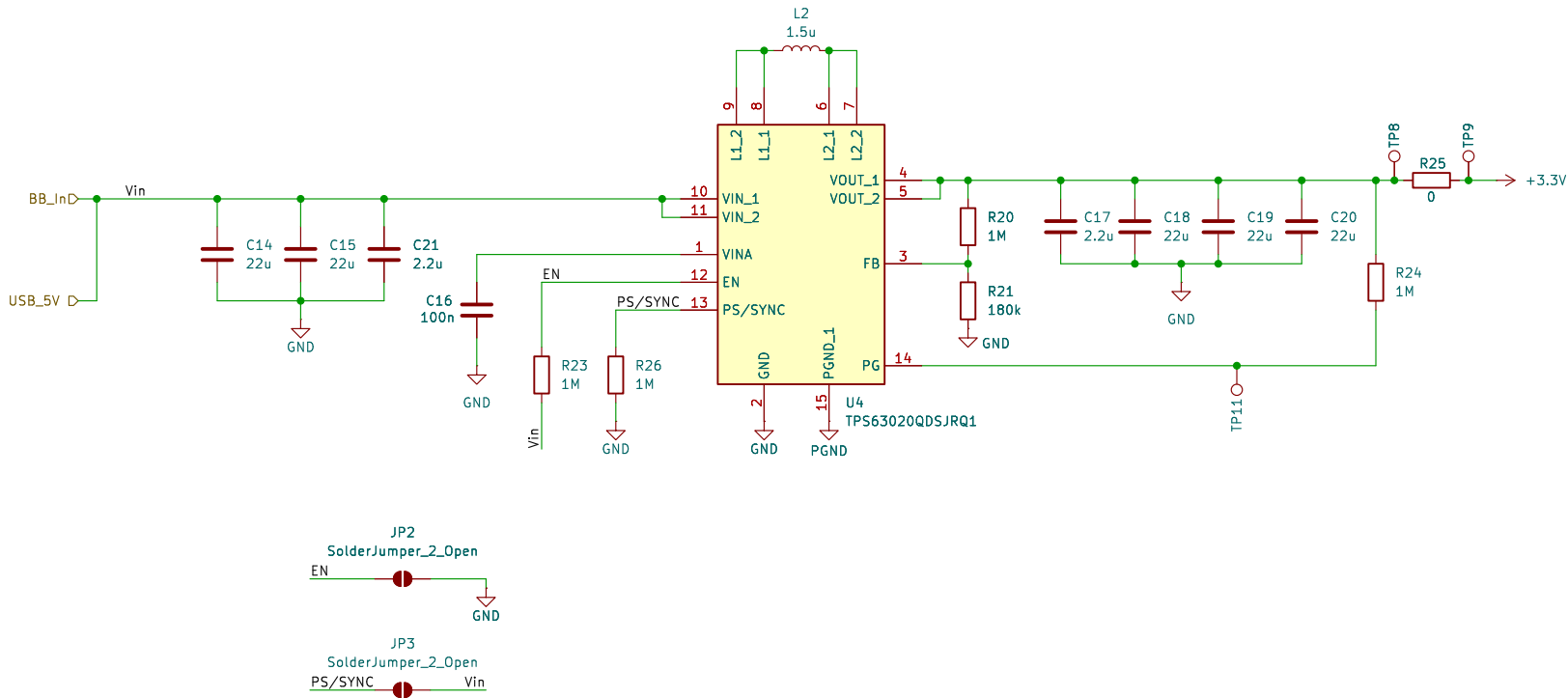
Rev:

Id: 2/12



Sheet: /Power Subsystem/	
File: power-subsystem.kicad_sch	
Title:	
Size: A4	Date:
KiCad E.D.A. 9.0.6	Rev:
	Id: 3/12

C: DC-DC Conversion (Voltage Regulation)



Sheet: /Power Subsystem/Regulators/
File: Other Blocks.kicad_sch

Title:

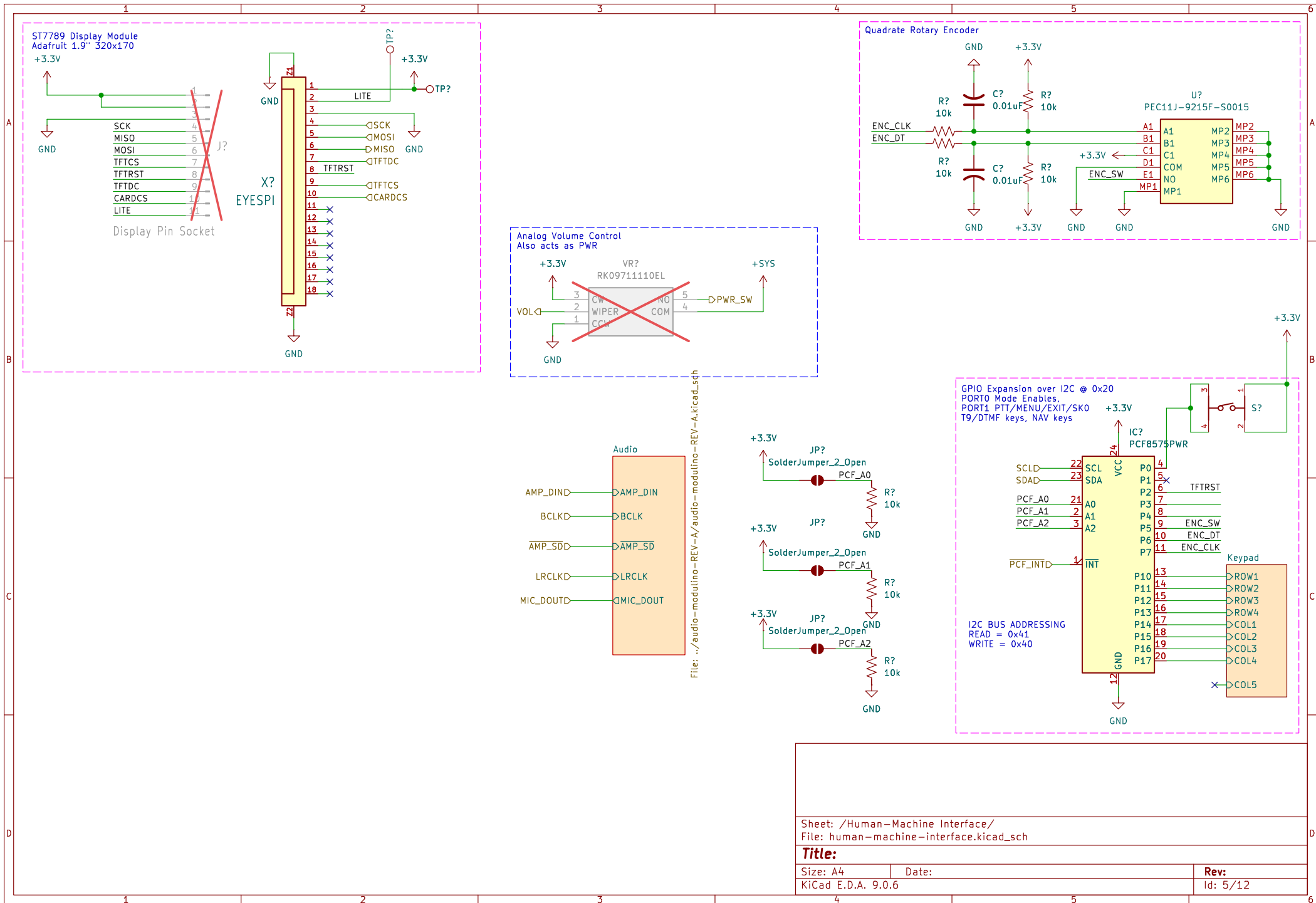
Size: B

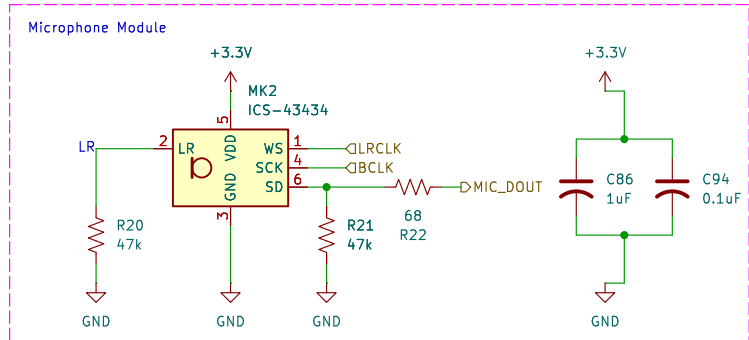
Date:

Rev:

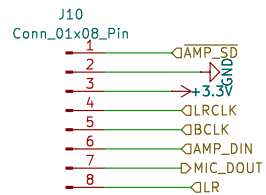
KiCad E.D.A. 9.0.6

Id: 4/12

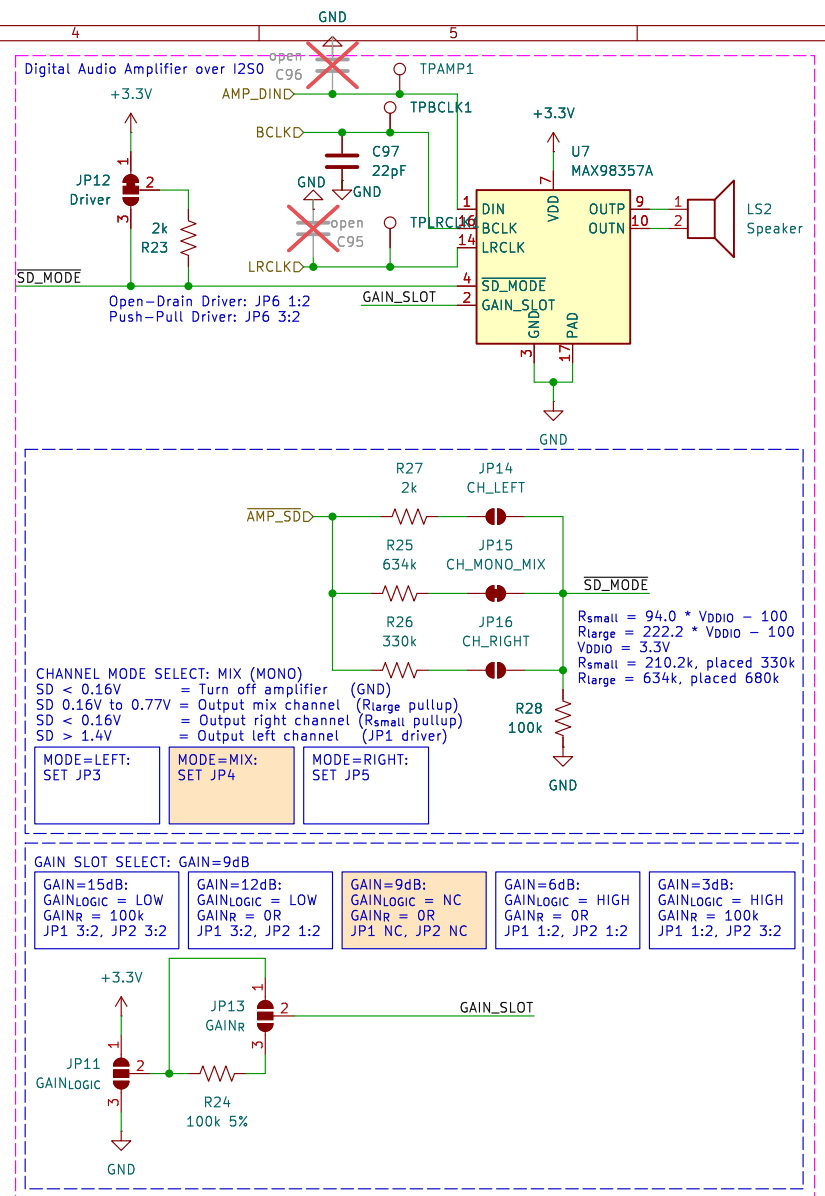




BreadboardConnector



LR-In case of two micro modulino



Sheet: /Human-Machine Interface/Audio/
File: audio-modulino-REV-A.kicad_sch

Title:

Size: A4

Date:

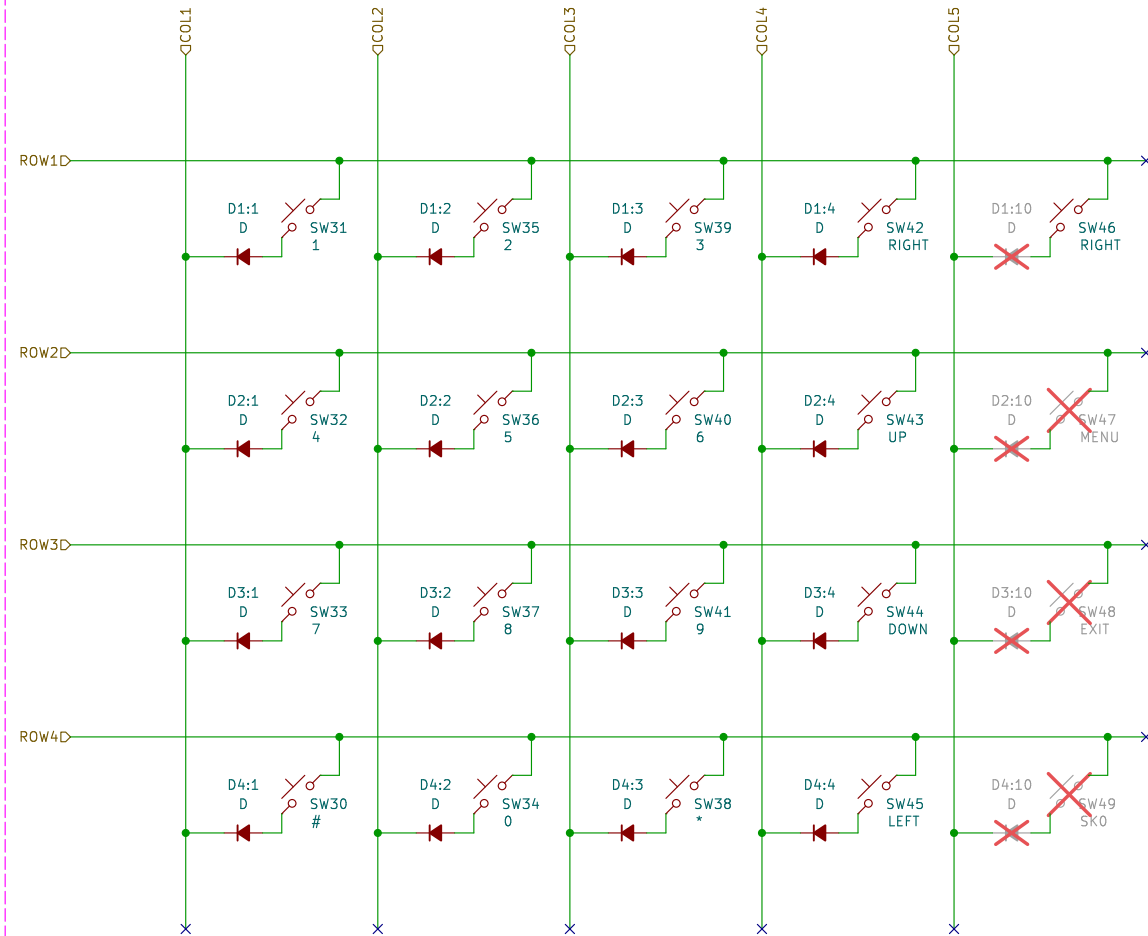
KiCad E.D.A. 9.0.6

Rev:

Id: 6/12

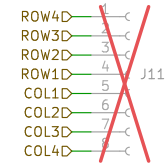
Keypad Matrix

Application Note: The ROW pins get pulled up by the GPIO expander, with one ROW being pulled down.
The scanner relies on the COL pins being read in as a logic swing to low.
Remark: as Domino Logic, a rail-to-rail logic swing occurs.



Keypad Daughterboard

HW-B34 Module



Sheet: /Human-Machine Interface/Keypad/
File: keypad.kicad_sch

Title:

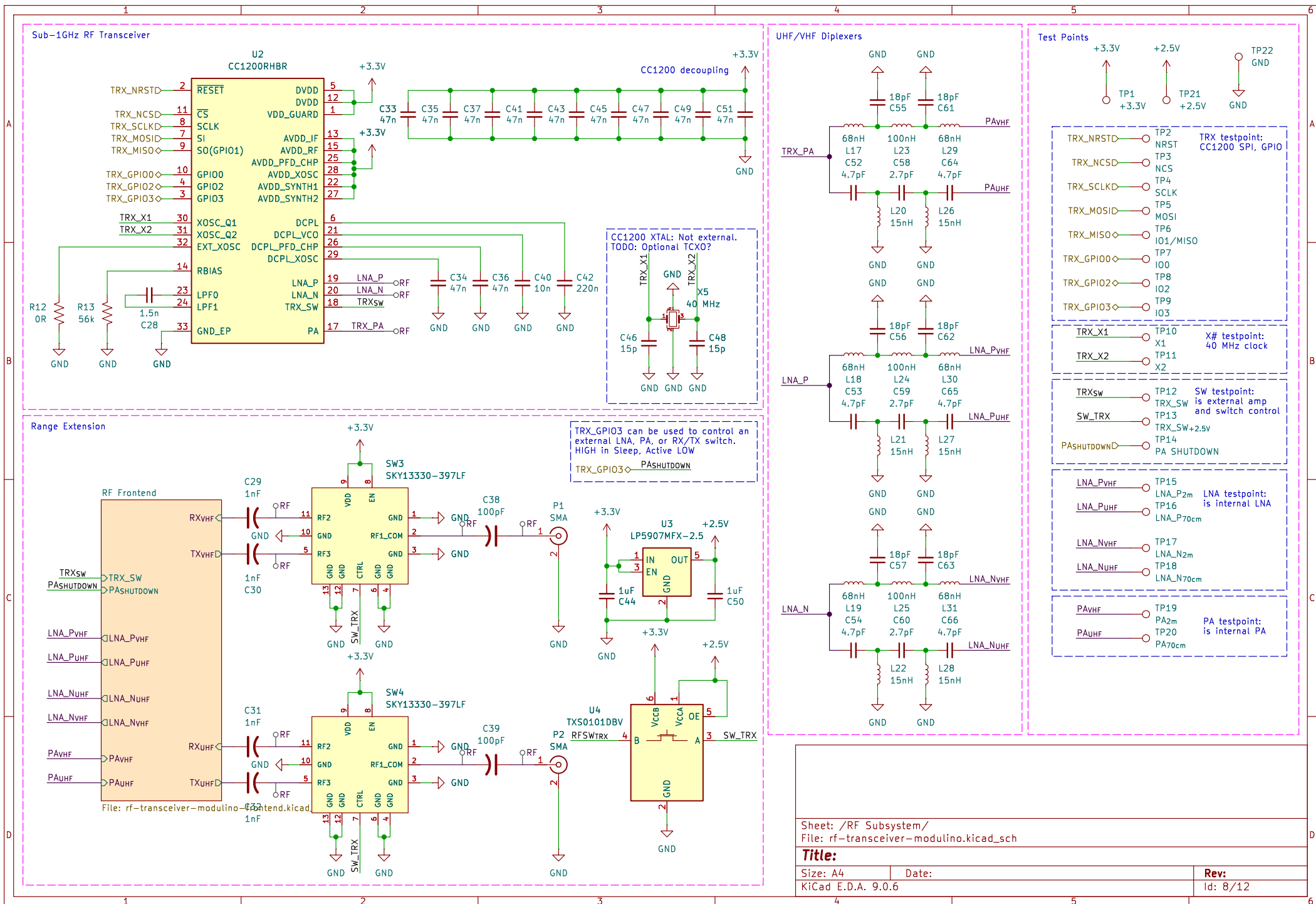
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Date:

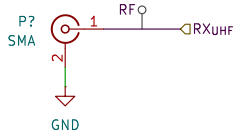
KiCad E.D.A. 9.0.6

Rev:

Id: 7/12



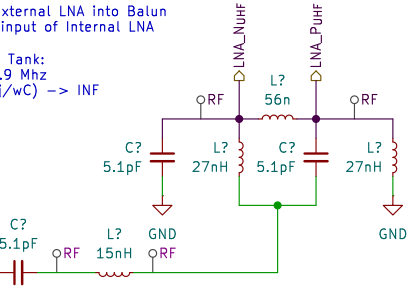
RF Frontend for UHF Reference 420-470 MHz Tune Separated RX and TX Signal Chains



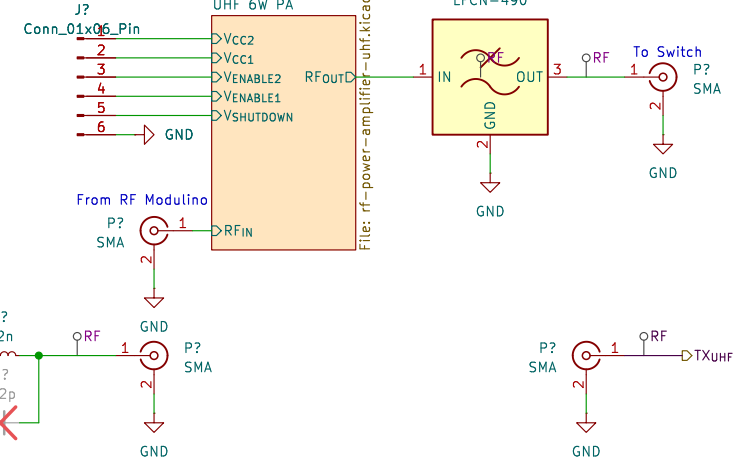
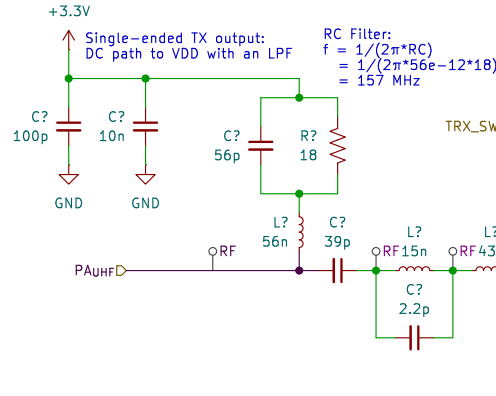
RX Signal Chain

Single-ended External LNA into Balun for differential input of Internal LNA

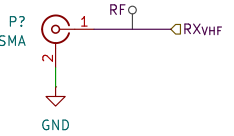
Balun as an LC Tank:
HIZ at $f = 429.9$ Mhz
(wL/wC)/($jwL-j/wC$) -> INF



TX Signal Chain

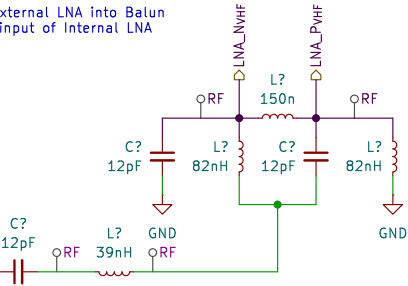


RF Frontend for VHF Reference 169-170 MHz Tune Separated RX and TX Signal Chains

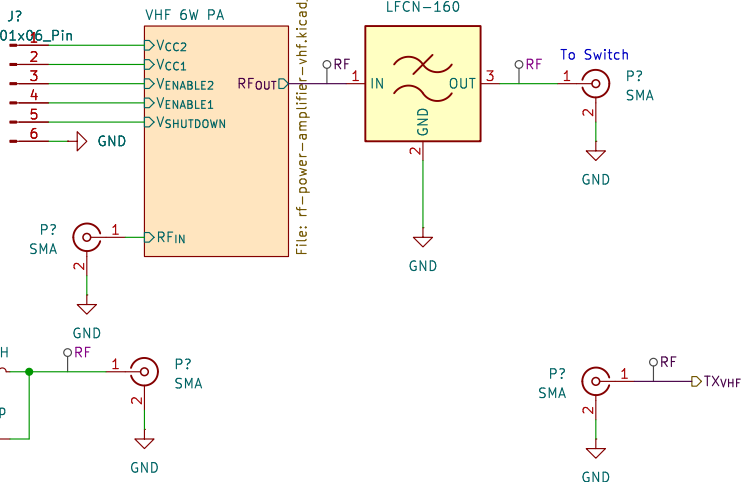
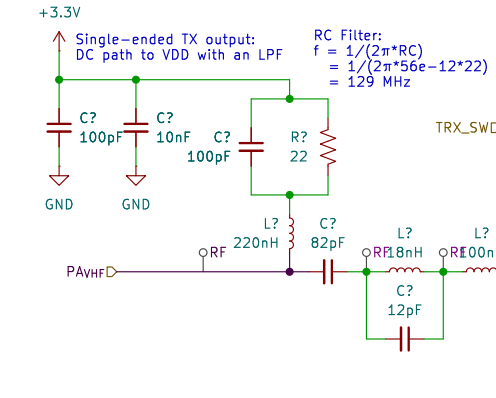


RX Signal Chain

Single-ended External LNA into Balun for differential input of Internal LNA



TX Signal Chain



Sheet: /RF Subsystem/RF Frontend/
File: rf-transceiver-modulino-frontend.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.6

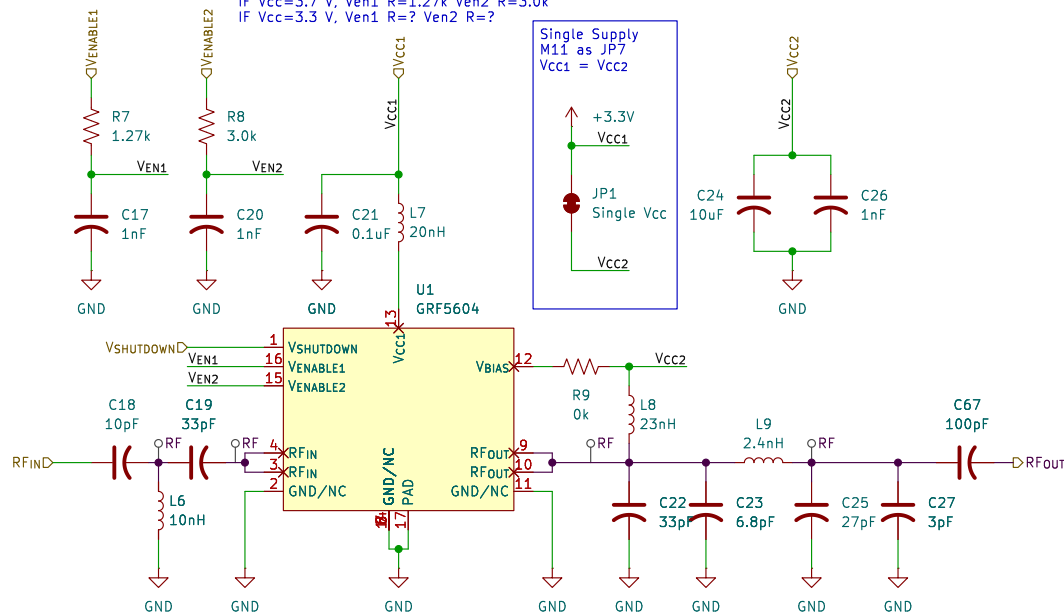
Rev:

Id: 9/12

RF Amplifier for UHF Reference 397-470 MHz Tune

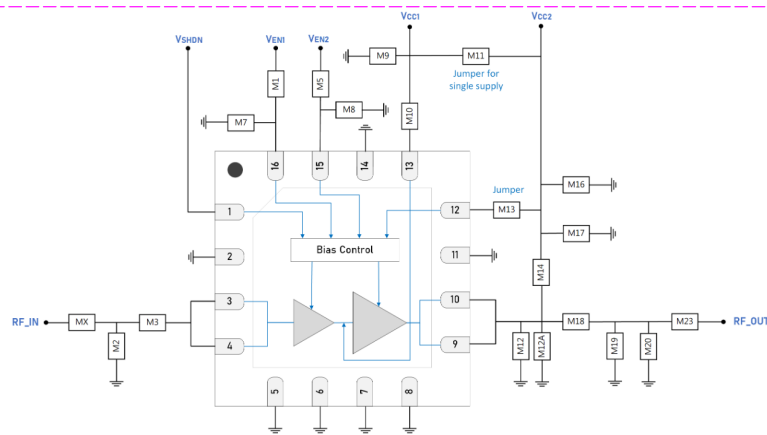
IF Vcc=5.0 V, Ven1 R=1.96k Ven2 R=6.04k
IF Vcc=3.7 V, Ven1 R=1.27k Ven2 R=3.0k
IF Vcc=3.3 V, Ven1 R=? Ven2 R=?

Single Supply
M11 as JP7
Vcc1 = Vcc2



Ref	Eval. Ref	450-470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 uF must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

Sheet: /RF Subsystem/RF Frontend/VHF 6W PA/
File: rf-power-amplifier-vhf.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.6

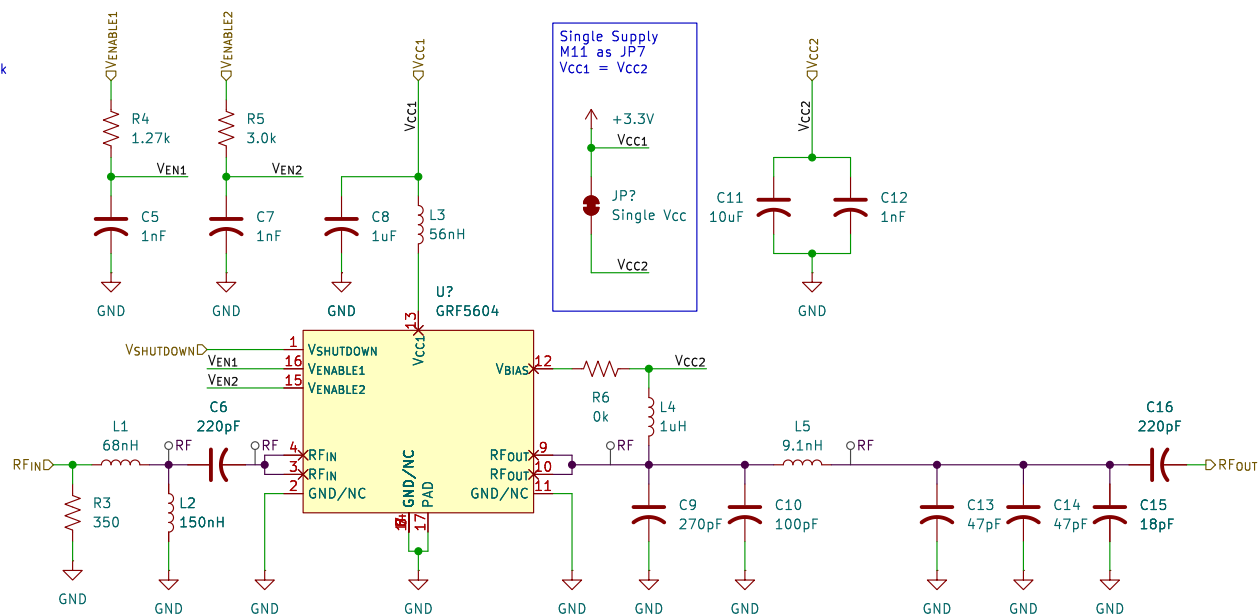
Date:

Rev:

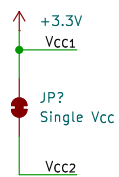
Id: 6/12

RF Amplifier for VHF Reference 135-175 MHz Tune

IF Vcc=5.0 V, Ven1 R=1.96k Ven2 R=6.04k
IF Vcc=3.7 V, Ven1 R=1.27k Ven2 R=3.0k
IF Vcc=3.3 V, Ven1 R=? Ven2 R=?

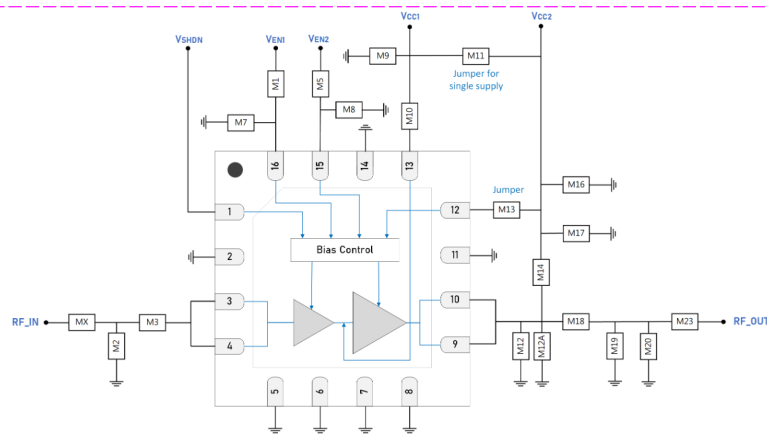


Single Supply
M11 as JP7
Vcc1 = Vcc2



Ref	Eval. Ref	450-470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 uF must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

Sheet: /RF Subsystem/RF Frontend/UHF 6W PA/
File: rf-power-amplifier-uhf.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 10/12